Chinook MySQL Analysis Report

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This report summarises the analysis of the Chinook dataset. The objective of this project was to extract and transform structured data, write efficient queries and solve business-driven questions.

The dataset consists of multiple related tables representing a digital music store, including customer, invoice, track, artist, genre, and playlist data. During the exploration phase, all tables were inspected using schema review and sample queries. This helped confirm table structures, column types, and relationships between tables, such as customer → invoice → invoiceline → track. This step ensured I had a clear understanding of how the various tables were related to one another before analysis began.

Missing data analysis was performed across key tables to assess data completeness. The customer and invoice tables had 29 and 202 missing values in the state and billing state columns, respectively. These missing values were in optional geographic fields and did not affect core transactional data. Track and invoiceline tables were also checked for missing values in key analytical fields such as TrackId, InvoiceId, UnitPrice, and Quantity. The results showed there were no missing data and the dataset was fully reliable.

To ensure accurate analysis, the relationship integrity between tables was tested using LEFT JOIN queries. All invoices were checked against customers to confirm valid customer links, Invoice lines were validated against invoices to ensure all sales transactions were properly recorded, and Track references in invoicelines were also validated.

Total revenue generated from invoices was 2328.60, representing the overall sales performance of the music store during the observed period. Revenue trends over time remained stable, with yearly revenues of 449.46 in 2021, 481.45 in 2022, 469.58 in 2023, 477.53 in 2024, and 450.58 in 2025. This consistency suggests steady customer demand and stable business performance across the years, with 2022 recording the highest revenue. The average order value was 5.65, indicating that customers typically made low- to moderate-priced purchases per transaction.

Customer analysis revealed that the top customer contributed 49.62 in total spending. Genre and track analysis showed that Rock was the highest performing category, while top tracks generated 3.98 in revenue. Geographic analysis revealed that the U.S was the highest revenue-generating region.

The analysis confirms that the dataset is structurally consistent, with minor missing values in non-critical fields. Core transactional relationships are intact, enabling reliable business analysis. Revenue is concentrated among specific customers, genres, and regions, indicating clear patterns in customer behaviour and product demand.

This project was completed using MySQL Workbench and Power BI.